

Activity__Explore sampling

May 12, 2026

1 Activity: Explore sampling

1.1 Introduction

In this activity, you will engage in effective sampling of a dataset in order to make it easier to analyze. As a data professional you will often work with extremely large datasets, and utilizing proper sampling techniques helps you improve your efficiency in this work.

For this activity, you are a member of an analytics team for the Environmental Protection Agency. You are assigned to analyze data on air quality with respect to carbon monoxide—a major air pollutant—and report your findings. The data utilized in this activity includes information from over 200 sites, identified by their state name, county name, city name, and local site name. You will use effective sampling within this dataset.

1.2 Step 1: Imports

1.2.1 Import packages

Import pandas, numpy, matplotlib, statsmodels, and scipy.

```
[1]: # Import libraries and packages

### YOUR CODE HERE ###
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats
import statsmodels.api as sm
```

1.2.2 Load the dataset

As shown in this cell, the dataset has been automatically loaded in for you. You do not need to download the .csv file, or provide more code, in order to access the dataset and proceed with this lab. Please continue with this activity by completing the following instructions.

```
[2]: # RUN THIS CELL TO IMPORT YOUR DATA.

### YOUR CODE HERE ###
epa_data = pd.read_csv("c4_epa_air_quality.csv", index_col = 0)
```

Hint 1

Use the function in the `pandas` library that allows you to read in data from a csv file and load it into a DataFrame.

Hint 2

Use the `read_csv` function from the `pandas` library. Set the `index_col` parameter to 0 to read in the first column as an index (and to avoid "Unnamed: 0" appearing as a column in the resulting Dataframe).

1.3 Step 2: Data exploration

1.3.1 Examine the data

To understand how the dataset is structured, examine the first 10 rows of the data.

```
[3]: # First 10 rows of the data

### YOUR CODE HERE ###
epa_data.head(10)
```

```
[3]:
```

	date_local	state_name	county_name	city_name \
0	2018-01-01	Arizona	Maricopa	Buckeye
1	2018-01-01	Ohio	Belmont	Shadyside
2	2018-01-01	Wyoming	Teton	Not in a city
3	2018-01-01	Pennsylvania	Philadelphia	Philadelphia
4	2018-01-01	Iowa	Polk	Des Moines
5	2018-01-01	Hawaii	Honolulu	Not in a city
6	2018-01-01	Hawaii	Honolulu	Not in a city
7	2018-01-01	Pennsylvania	Erie	Erie
8	2018-01-01	Hawaii	Honolulu	Honolulu
9	2018-01-01	Colorado	Larimer	Fort Collins

	local_site_name	parameter_name \
0	BUCKEYE	Carbon monoxide
1	Shadyside	Carbon monoxide
2	Yellowstone National Park - Old Faithful Snow ...	Carbon monoxide
3	North East Waste (NEW)	Carbon monoxide
4	CARPENTER	Carbon monoxide
5	Kapolei	Carbon monoxide
6	Kapolei	Carbon monoxide
7	NaN	Carbon monoxide

```

8                                     Honolulu Carbon monoxide
9                                Fort Collins - CSU - S. Mason Carbon monoxide

```

```

      units_of_measure  arithmetic_mean  aqi
0  Parts per million      0.473684      7
1  Parts per million      0.263158      5
2  Parts per million      0.111111      2
3  Parts per million      0.300000      3
4  Parts per million      0.215789      3
5  Parts per million      0.994737     14
6  Parts per million      0.200000      2
7  Parts per million      0.200000      2
8  Parts per million      0.400000      5
9  Parts per million      0.300000      6

```

Hint 1

Use the function in the `pandas` library that allows you to get a specific number of rows from the top of a DataFrame.

Hint 2

Use the `head` function from the `pandas` library. Set the `n` parameter to 10 to print out the first 10 rows.

Question: What does the `aqi` column represent?

The `aqi` column represents the Air Quality Index for the districts.

1.3.2 Generate a table of descriptive statistics

Generate a table of some descriptive statistics about the data. Specify that all columns of the input be included in the output.

```

[5]: ### YOUR CODE HERE ###
      epa_data.describe(include='all')

```

```

[5]:   date_local  state_name  county_name  city_name  local_site_name  \
count      260         260         260         260             257
unique        1          52         149         190             253
top    2018-01-01  California  Los Angeles  Not in a city      Kapolei
freq        260          66          14          21              2
mean         NaN          NaN          NaN          NaN          NaN
std          NaN          NaN          NaN          NaN          NaN
min          NaN          NaN          NaN          NaN          NaN
25%          NaN          NaN          NaN          NaN          NaN
50%          NaN          NaN          NaN          NaN          NaN
75%          NaN          NaN          NaN          NaN          NaN
max          NaN          NaN          NaN          NaN          NaN

```

	parameter_name	units_of_measure	arithmetic_mean	aqi
count	260	260	260.000000	260.000000
unique	1	1	NaN	NaN
top	Carbon monoxide	Parts per million	NaN	NaN
freq	260	260	NaN	NaN
mean	NaN	NaN	0.403169	6.757692
std	NaN	NaN	0.317902	7.061707
min	NaN	NaN	0.000000	0.000000
25%	NaN	NaN	0.200000	2.000000
50%	NaN	NaN	0.276315	5.000000
75%	NaN	NaN	0.516009	9.000000
max	NaN	NaN	1.921053	50.000000

Hint 1

Use function in the **pandas** library that allows you to generate a table of basic descriptive statistics in a DataFrame.

Hint 2

Use the **describe** function from the **pandas** library. Set the **include** parameter passed in to this function to 'all' to specify that all columns of the input be included in the output.

Question: Based on the preceding table of descriptive statistics, what is the mean value of the **aqi** column?

Based on the preceding table of descriptive statistics, the mean value of the **aqi** column is 6.757692.

Question: Based on the preceding table of descriptive statistics, what do you notice about the count value for the **aqi** column?

Based on the preceding table of descriptive statistics, the count for **aqi** column is 260. That means there are 260 AQI measurements in the dataset.

1.3.3 Use the `mean()` function on the **aqi** column

Now, use the `mean()` function on the **aqi** column and assign the value to a variable **population_mean**. The value should be the same as the one generated by the `describe()` method in the above table.

```
[7]: ### YOUR CODE HERE ###
population_mean = epa_data['aqi'].mean()
population_mean
```

```
[7]: 6.757692307692308
```

Hint 1

Use the function in the **pandas** library that allows you to generate a mean value for a column in a DataFrame.

Hint 2

Use the `mean()` method.

1.4 Step 3: Statistical tests

1.4.1 Sample with replacement

First, name a new variable `sampled_data`. Then, use the `sample()` dataframe method to draw 50 samples from `epa_data`. Set `replace` equal to `'True'` to specify sampling with replacement. For `random_state`, choose an arbitrary number for random seed. Make that arbitrary number 42.

```
[8]: ### YOUR CODE HERE ###
sampled_data = epa_data.sample(n=50, replace=True, random_state=42)
```

1.4.2 Output the first 10 rows

Output the first 10 rows of the DataFrame.

```
[9]: ### YOUR CODE HERE ###
sampled_data.head(10)
```

```
[9]:
```

	date_local	state_name	county_name	city_name	\
102	2018-01-01	Texas	Harris	Houston	
106	2018-01-01	California	Imperial	Calexico	
71	2018-01-01	Alabama	Jefferson	Birmingham	
188	2018-01-01	Arizona	Maricopa	Tempe	
20	2018-01-01	Virginia	Roanoke	Vinton	
102	2018-01-01	Texas	Harris	Houston	
121	2018-01-01	North Carolina	Mecklenburg	Charlotte	
214	2018-01-01	Florida	Broward	Davie	
87	2018-01-01	California	Humboldt	Eureka	
99	2018-01-01	California	Santa Barbara	Goleta	

	local_site_name	parameter_name	units_of_measure	\
102	Clinton	Carbon monoxide	Parts per million	
106	Calexico-Ethel Street	Carbon monoxide	Parts per million	
71	Arkadelphia/Near Road	Carbon monoxide	Parts per million	
188	Diablo	Carbon monoxide	Parts per million	
20	East Vinton Elementary School	Carbon monoxide	Parts per million	
102	Clinton	Carbon monoxide	Parts per million	
121	Garinger High School	Carbon monoxide	Parts per million	
214	Daniela Banu NCORE	Carbon monoxide	Parts per million	
87	Jacobs	Carbon monoxide	Parts per million	
99	Goleta	Carbon monoxide	Parts per million	


```
arithmetic_mean aqi
```

102	0.157895	2
106	1.183333	26
71	0.200000	2
188	0.542105	10
20	0.100000	1
102	0.157895	2
121	0.200000	2
214	0.273684	5
87	0.393750	5
99	0.222222	3

Hint 1

Use the function in the `pandas` library that allows you to get a specific number of rows from the top of a `DataFrame`.

Hint 2

Use the `head` function from the `pandas` library. Set the `n` parameter to 10 to print out the first 10 rows.

Question: In the `DataFrame` output, why is the row index 102 repeated twice?

The row index 102 is repeated twice because we have ‘True’ for `replace` argument, allowing sampling units to occur more than once. In this case, row 102 happened to occur more than once.

Question: What does `random_state` do?

`random_state` assigns a random seed number to generate a random sample which allows for the reproduction of the same exact sample each time the command is run.

1.4.3 Compute the mean value from the `aqi` column

Compute the mean value from the `aqi` column in `sampled_data` and assign the value to the variable `sample_mean`.

```
[10]: ### YOUR CODE HERE ###
      sample_mean = sampled_data['aqi'].mean()
      sample_mean
```

```
[10]: 5.54
```

Question: Why is `sample_mean` different from `population_mean`?

`sample_mean` is different from `population_mean` due to sampling variability.

1.4.4 Apply the central limit theorem

Imagine repeating the the earlier sample with replacement 10,000 times and obtaining 10,000 point estimates of the mean. In other words, imagine taking 10,000 random samples of 50 AQI values

and computing the mean for each sample. According to the **central limit theorem**, the mean of a sampling distribution should be roughly equal to the population mean. Complete the following steps to compute the mean of the sampling distribution with 10,000 samples.

- Create an empty list and assign it to a variable called `estimate_list`.
- Iterate through a `for` loop 10,000 times. To do this, make sure to utilize the `range()` function to generate a sequence of numbers from 0 to 9,999.
- In each iteration of the loop, use the `sample()` function to take a random sample (with replacement) of 50 AQI values from the population. Do not set `random_state` to a value.
- Use the list `append()` function to add the value of the sample mean to each item in the list.

```
[11]: ### YOUR CODE HERE ###

estimate_list = []
for i in range(10000):
    estimate_list.append(epa_data['aqi'].sample(n=50,replace=True).mean())
```

Hint 1

Review [the content about sampling in Python](#).

1.4.5 Create a new DataFrame

Next, create a new DataFrame from the list of 10,000 estimates. Name the new variable `estimate_df`.

```
[12]: ### YOUR CODE HERE ###

estimate_df = pd.DataFrame(data={'estimate': estimate_list})
estimate_df
```

```
[12]:      estimate
0         9.94
1         6.06
2         6.86
3         7.34
4         6.64
...      ...
9995      6.00
9996      6.80
9997      7.54
9998      6.50
9999      6.48
```

```
[10000 rows x 1 columns]
```

Hint 1

Review [the content about sampling in Python](#).

Hint 2

Use the `mean()` function.

1.4.6 Compute the mean() of the sampling distribution

Next, compute the `mean()` of the sampling distribution of 10,000 random samples and store the result in a new variable `mean_sample_means`.

```
[13]: ### YOUR CODE HERE ###

mean_sample_means = estimate_df['estimate'].mean()
mean_sample_means
```

```
[13]: 6.7684640000000265
```

Hint 1

Use the function in the `pandas` library that allows you to generate a mean value for a column in a `DataFrame`.

Hint 2

Use the `mean()` function.

Question: What is the mean for the sampling distribution of 10,000 random samples?

This number will vary as `random_state` was not set to a value.

Hint 3

This value is contained in `mean_sample_means`.

Hint 4

According to the central limit theorem, the mean of the preceding sampling distribution should be roughly equal to the population mean.

Question: How are the central limit theorem and random sampling (with replacement) related?

Central limit theorem relates to random sampling because it states that if the sample size is large enough and the observations are drawn independently, the sampling distribution of the sample mean is approximately a normal distribution.

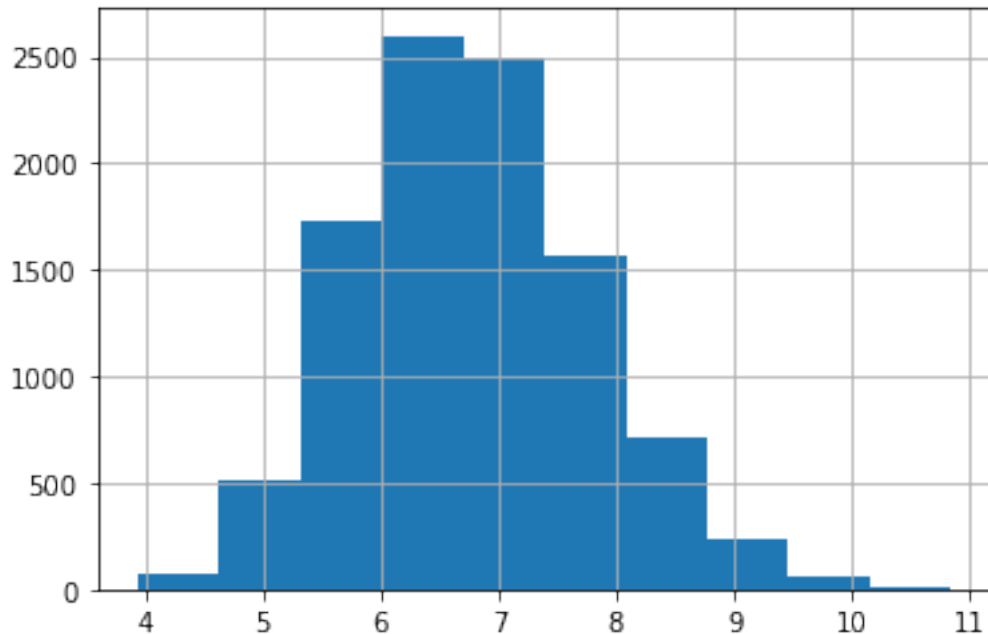
1.4.7 Output the distribution using a histogram

Output the distribution of these estimates using a histogram. This provides an idea of the sampling distribution.

```
[14]: ### YOUR CODE HERE ###

estimate_df['estimate'].hist()
```

```
[14]: <matplotlib.axes._subplots.AxesSubplot at 0x717fc0aa7710>
```

Hint 1

Use the `hist()` function.

1.4.8 Calculate the standard error

Calculate the standard error of the mean AQI using the initial sample of 50. The **standard error** of a statistic measures the sample-to-sample variability of the sample statistic. It provides a numerical measure of sampling variability and answers the question: How far is a statistic based on one particular sample from the actual value of the statistic?

```
[15]: ### YOUR CODE HERE ###

standard_error = sampled_data['aqi'].std() / np.sqrt(len(sampled_data))
standard_error
```

```
[15]: 0.7413225908290327
```

Hint 1

Use the `std()` function and the `np.sqrt()` function.

1.5 Step 4: Results and evaluation

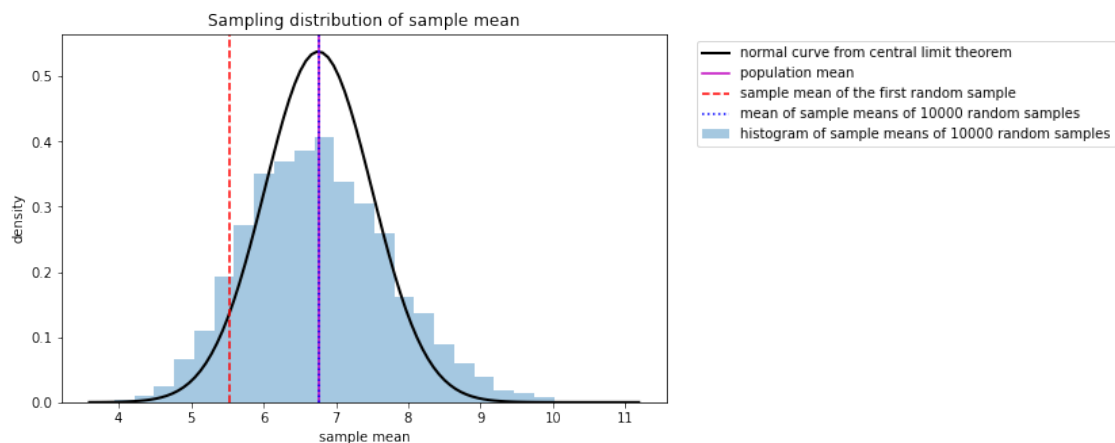
1.5.1 Visualize the relationship between the sampling and normal distributions

Visualize the relationship between your sampling distribution of 10,000 estimates and the normal distribution.

1. Plot a histogram of the 10,000 sample means
2. Add a vertical line indicating the mean of the first single sample of 50
3. Add another vertical line indicating the mean of the means of the 10,000 samples
4. Add a third vertical line indicating the mean of the actual population

```
[17]: ### YOUR CODE HERE ###

plt.figure(figsize=(8,5))
plt.hist(estimate_df['estimate'], bins=25, density=True, alpha=0.4, label =
    ↪ "histogram of sample means of 10000 random samples")
xmin, xmax = plt.xlim()
x = np.linspace(xmin, xmax, 100) # generate a grid of 100 values from xmin to
    ↪ xmax.
p = stats.norm.pdf(x, population_mean, standard_error)
plt.plot(x, p, 'k', linewidth=2, label = 'normal curve from central limit
    ↪ theorem')
plt.axvline(x=population_mean, color='m', linestyle = 'solid', label =
    ↪ 'population mean')
plt.axvline(x=sample_mean, color='r', linestyle = '--', label = 'sample mean of
    ↪ the first random sample')
plt.axvline(x=mean_sample_means, color='b', linestyle = ':', label = 'mean of
    ↪ sample means of 10000 random samples')
plt.title("Sampling distribution of sample mean")
plt.xlabel('sample mean')
plt.ylabel('density')
plt.legend(bbox_to_anchor=(1.04,1));
```



Question: What insights did you gain from the preceding sampling distribution?

1. The histogram of the sampling distribution closely mirrors a normal histogram.
2. Due to sampling variability, the estimate based on one particular sample is off-center.
3. The population mean and the mean of the sample means overlap, meaning they are almost the same.

2 Considerations

What are some key takeaways that you learned from this lab? Sampling with replacement on a dataset leads to duplicate rows. Sample means are different from population means due to sampling variability. The central limit theorem helps describe the sampling distribution of the sample mean for many different types of datasets.

What findings would you share with others? The mean AQI in a sample of 50 observations was below 100) meaning the AQI is under satisfactory levels. However, there are some districts that have higher AQI levels than others which need more attention.

What would you convey to external stakeholders? Comparatively more funding should be allocated to further investigate regions with unhealthy levels of carbon monoxide and to improve the conditions in those regions.

Congratulations! You've completed this lab. However, you may not notice a green check mark next to this item on Coursera's platform. Please continue your progress regardless of the check mark. Just click on the "save" icon at the top of this notebook to ensure your work has been logged.